

ZILONG ZHANG

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EDUCATION

Ph.D. in Machine Learning & Data Science

Aug. 2025 – Now

Department of Mathematics & Statistics / Georgia State University

Atlanta, GA

Master in Statistics

Aug. 2023 – May. 2025

Department of Statistics & Probability / Michigan State University

East Lansing, MI

Bachelor of Financial Mathematics

Sep. 2019 – Jul. 2023

School of Statistics / University of International Business and Economics

Beijing, China

RESEARCH INTERESTS

- **Statistical Machine Learning:** graph neural networks, transfer learning, and statistical theory of deep learning
- **Functional Data Analysis (FDA):** time series, functional principal component analysis, bandwidth selection
- **Bioinformatics / Computational Biology:** single-cell multi-omics integration, RNA–protein prediction, and molecular data analysis

PUBLICATIONS & MANUSCRIPTS

- Xuan Kan, Zimu Li, Hejie Cui, Yue Yu, Ran Xu, Shaojun Yu, **Zilong Zhang**, Ying Guo, Carl Yang. “R-Mixup: Riemannian Mixup for Biological Networks” *Proceedings of SIGKDD Conference on Knowledge Discovery and Data Mining 2023*, **SIGKDD 2023**.
- Xuan Kan, Wei Dai, Hejie Cui, **Zilong Zhang**, Ying Guo, Carl Yang. “Brain Network Transformer” *Proceedings of the Conference on Neural Information Processing Systems 2022*, **NeurIPS 2022**.

RESEARCH AND OTHER PROJECTS

mRNA to Protein Prediction based on Transformer and GNN

Jan. 2024 – Jan. 2026

- *Project Leaders:* Dr. Yuying Xie (Michigan State University) and Dr. Lulu Shang (MD Anderson Cancer Center)
- Developed a novel predictive model based on Transformer and Graph Neural Networks (GNN) to accurately predict protein expression from single-cell RNA data

CRIC Study: Genome-Wide Association of Apparent Treatment-Resistant Hypertension in CKD and ARTH Patients

Sept. 2024 – Dec. 2024

- *Project leader:* Dr. Ling Wang (Michigan State University) and Dr. Yuehua Cui (Michigan State University)
- Analyzed high-dimensional genetic data to identify key genetic variants that contribute to treatment-resistant hypertension in CKD patients, providing potential targets for therapeutic intervention.

Theoretical Research and Application of High Dimensional Variable Screening

Oct. 2022 – May. 2023

- *Project leader:* Dr. Wei Xiong (University of International Business and Economics)
- Combined the application of high-dimensional point map mapping in brain map neural network, to find the most efficient dimensionality reduction method in a specific neural networks situation according to univariate analysis screening method and multivariate analysis screening method.

R-Mixup: Riemannian Mixup for Biological Networks

Jun. 2022 – Mar. 2023

- *Project leader:* Dr. Carl Yang (Emory University)
- Participated in designing our optimization model in neural network, and Complete the regression analysis part of the generalization of a specific model of neural network to Riemannian manifolds.

Brain Network Transformer

Jan. 2022 – Sep. 2022

- *Project leader:* Dr. Carl Yang (Emory University)
- Considered process samples by a general nonlinear regression with cluster centers as variables, compared the coefficient of determination between the orthonormal and nonorthonormal cases using the concept of Regression Analysis, and assured the conclusion through the innovational use of significance level in hypothesis testing.

WORK EXPERIENCE

China Securities Co., Ltd.

Jul. 2021 – Nov. 2021

Intern Investment Manager

Beijing, China

- Assisted in listing work of Beijing Tianzhilang Medical Technology Co., LTD and Golden Sun Grain and Oil Co., LTD.
- Prepared listing application materials, such as industry survey reports and risk assessment reports.
- Planned and executed fundraising activities. Composed market research reports and other review materials.

Chuzhou Longyun Financial Consulting Co. Ltd.

Mar. 2021 – Jun. 2021

General Manager Assistant

Chuzhou, China

- Provided regular financial audit and accounting services to clients.
- Conducted corporate financial audit and due diligence for the corporate.

Bank of Communications Chuzhou Branch

Dec. 2020 – Feb. 2021

Intern Account Manager

Chuzhou, China

- Identified, managed and mitigated accounting risk.
- Set up and maintained credit databases to carry out related creditability assessments and analysis reports.

TECHNICAL SKILLS

Programming: Python, R, C, Sklearn, Numpy, Pandas, PyTorch, Tensorflow, Huggingface

Tools: Git, Linux, Jupyter, VS Code

Skills: Large Language Models, Trustworthy AI, Data Analysis, Statistical Learning Theory

HONORS & AWARDS

- **Second Prize**, Beijing Sub-Region of the 11th National University Students E-commerce “Innovation, Creativity and Entrepreneurship” Challenge Jun. 2021
- **Silver Award**, “Challenge Cup” Capital University Students Business Plan Competition Sep. 2020

PROFESSIONAL MEMBERSHIPS

- Student Member, **International Chinese Statistical Association(ICSA)** 2026–Present